

Supplementary Document

Sensitivity Analysis

To evaluate whether trend and change-point results are sensitive to the interpolation of missing GRACE/GRACE-FO data-availability gaps, Mann-Kendall trend test, Sen's Slope and Pettitt's homogeneity test were repeated on raw (missing values not addressed) GWSA series for October and November, across the study area. Over the 22-year study period, data for the month of November has gaps in the years of 2015 and 2017 and October month has missing data in 2012, 2015, 2016, and 2017.

The Linear interpolation had a minimal impact on the trend classification in October and November months (Fig. S1 and S2). The raw and interpolated datasets showed matching trend directions and significance across ~97% of grid cells, with a mean absolute change in Kendall's τ of just 0.08 in November. October month also showed similarity of 57%, and mean Kendall's τ decreased from 0.33 to 0.14, reflecting to the month's higher number of missing datasets.

The non-parametric trend analysis of post-monsoon GWSA reveals a consistent north-south spatial polarity across the Indian Peninsula. The pixels north of 15°N shows a universally positive Kendall's τ indicating widespread groundwater storage recovery. In contrast, the region south of 11°N (Tirunelveli and Virudhunagar districts) shows a clear negative trend, with 70–77% declining, pointing to localized long-term depletion. One additional cell, located separately at 13.25°N 80.25°E near Tiruvallur, shows a distinct and more extreme negative signal, an isolated decoupling signature rather than the broader southern depletion cluster.

For Pettitt's change-point test, exact-year agreement was similar in the two months (October: 51%; November: 50%) and the assigned breakpoint year matched within two years in 68.1% of cells of October and 52.8% of November (Fig. S3 and S4). Both showed a substantial share of cells with a ≥ 2 -year shift in the assigned breakpoint year under the two treatments (October: 31.9%; November: 47.2%). This indicates that while the trend/significance classification is comparatively stable in November, the precise year attributed to a regime shift by Pettitt's test should be treated as an approximate estimate rather than a fixed value, regardless of the interpolation approach used in both months.

Cross-referencing these results against the critical groundwater sites identified in southern Tamil Nadu and Kerala shows that this designation is substantively supported by the more robust November trend (lagged effect) signals. The 67% of the core critical cells retained a

significant negative Kendall's τ under both the raw and interpolated treatments in November, with considerably less contribution from the month of October.

Taken together, the sensitivity analysis confirms that the choice of gap-handling methods or addressing the missing values as it does not materially alter the study's primary conclusions. The critical area findings remain consistent in both trend direction and statistical significance across the missing dataset. The October month's trend classification shows higher sensitivity to missing-data treatment. The Pettitt's test change-point years for either month should be regarded as approximate rather than precise, independent of interpolation technique.

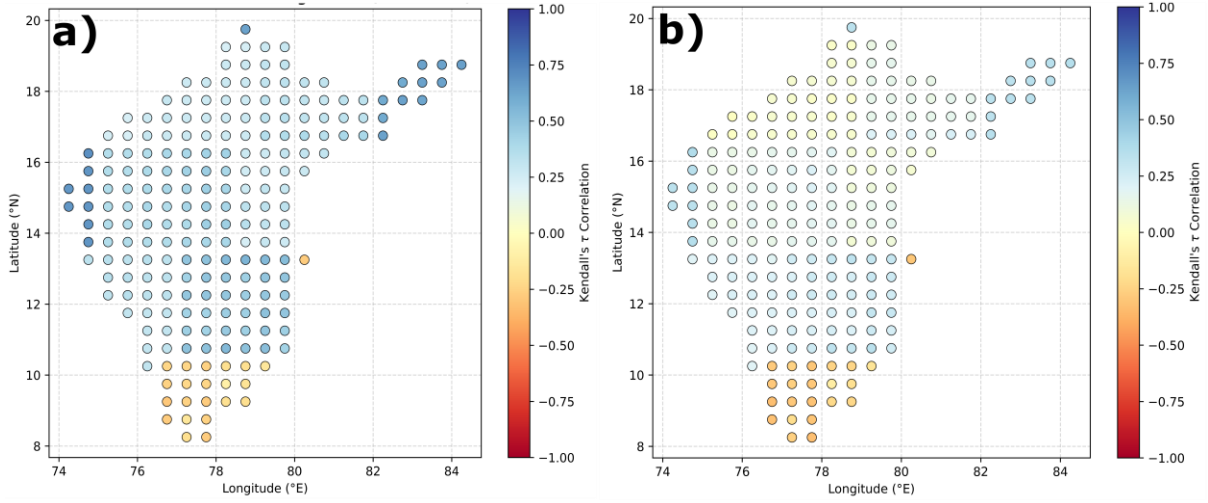


Fig. S1 Spatial distribution of post-monsoon groundwater anomaly trends of (a) missing and (b) interpolated data during October based on Kendall's tau values, indicating the strength and direction of groundwater change.

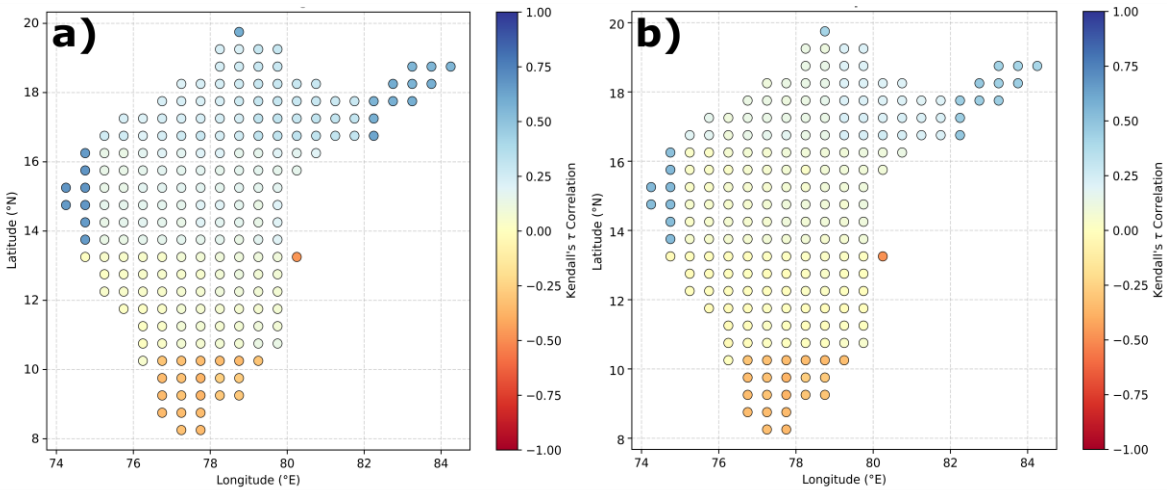


Fig. S2 Spatial distribution of post-monsoon groundwater anomaly trends of (a) missing and (b) interpolated data during November based on Kendall's tau values, indicating the strength and direction of groundwater change.

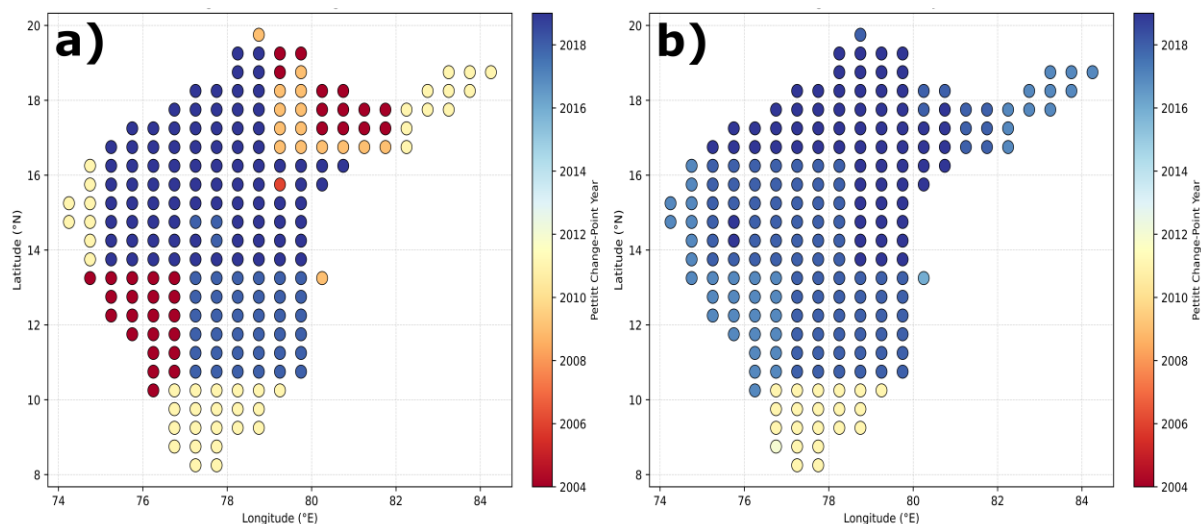


Fig. S3 Spatial distribution of significant breakpoints in post-monsoon groundwater anomaly homogeneity of (a) missing and (b) interpolated data during October as identified using Pettitt's test ($p < 0.05$).

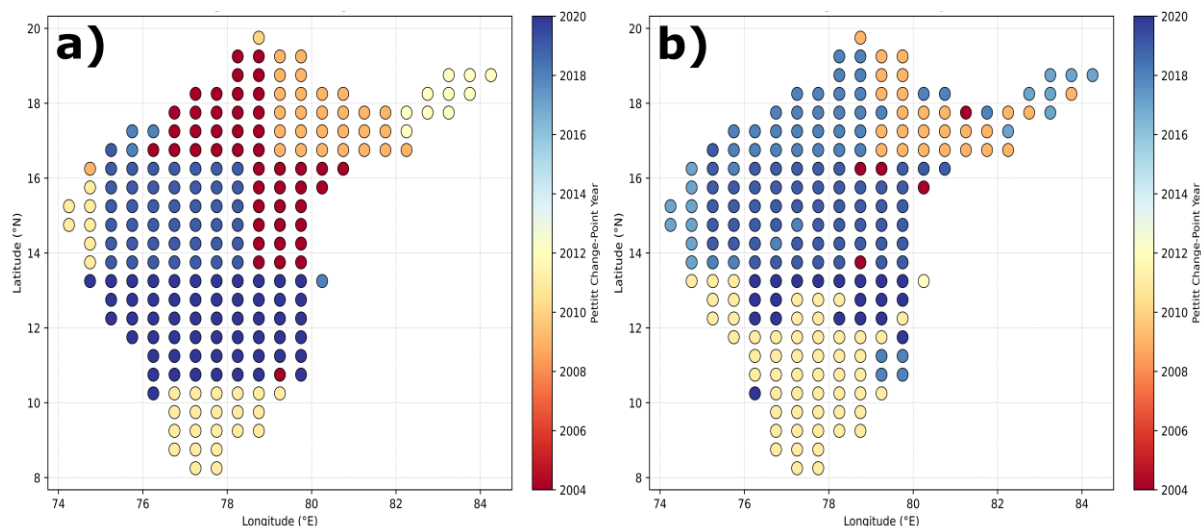


Fig. S4 Spatial distribution of significant breakpoints in post-monsoon groundwater anomaly homogeneity of (a) missing and (b) interpolated data during November as identified using Pettitt's test ($p < 0.05$).